HAN_UNIVERSITY
OF APPLIED SCIENCES

Deep Learning report

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By submitting this portfolio the authors certify that this is their original work, and they have cited all the referenced materials properly.

CONTENTS

1 Introduction 2
2 Problem statement 3
2.1 Problem statement 3
2.2 Goal 3
2.3 Requirements 3
2.3.1 Functional requirements 3
2.3.2 Non-functional requirements 4
3 Data augmentation and preprocessing 5
3.1 Dataset increasing 5
3.1.1 Key Details About the Dataset 5
3.1.2 Why Data Augmentation is Needed 5
3.1.3 How Data Augmentation Works 5
3.1.4 Implementation Steps 5
3.1.5 Benefits of Data Augmentation 5
3.1.6 Image Preprocessing Pipeline 5
3.2 Before/After Examples 6
3.3 Benefits and Outcome 6
4 CNN architecture, training and validation 7
4.1 Strategy 1: Manual CNN architecture 7
4.2 Strategy 2: AutoML solution with AutoKeras 8
4.3 Strategy 3: Keras Tuner 9
4.3.1 Decide number of layers 9
4.4 Model Tests 10
4.5 Layer visualization 12
4.6 Test Plan and Results 13
4.6.1 Functional Requirements Test Plan 13
4.6.2 Testplan Non-functional requirements 14
5 Conclusion 15
6 References 16
7 Code appendices 17
7.1 Data preprocessing 17
7.2 Model creation 18
7.3 Training process 20
7.4 Evaluation methods 21

1 Introduction

As part of the Embedded Vision and Machine Learning module, our portfolio will focus on deep learning (DL) applications for object identification and form recognition. Our project will be focused on hand gesture recognition. The Deep-learning course serves as an introduction to the tools that are commonly used such as Opencv, Keras and TensorFlow. While this program is a module rather than a minor for our study group, machine learning and deep learning plays a critical role in our studies. The course emphasizes hands-on applications of deep learning for computer vision. The focus on image processing and object classification aligns well with our project objectives, allowing us to develop practical solutions for real-world vision systems.

Our project group focuses on the role of deep learning in object recognition in images. With the increasing demand for intelligent vision systems across industries, Deep learning techniques for object recognition are essential in fields such as automation, robotics, and smart devices. The course's hands-on approach will help us build critical skills that can be applied to solve vision problems with deep learning in our future careers.

Our main project centers around using deep learning to identify gestures of the well-known game, rock paper scissors. By leveraging the computer vision techniques covered in the course, we aim to develop a system that efficiently detects these gestures. This practical application of ML is key to achieving our project's goals and deploying these models in real-world environments.

This document is a continuation of the previous machine learning report and specifically focuses on deep learning. The previous machine learning project will serve as the foundation for this project, where we will further explore and expand the concepts and techniques applied in the earlier project to more advanced deep learning methods. Deep learning, as a subset of machine learning, offers more powerful techniques for complex tasks such as object recognition and image processing. We will apply these techniques to improve and refine the performance of our system. The goal of this project is to create a deep learning model that detects efficiently the 3 hand gestures.

2 Problem statement

2.1 Problem statement

This project focuses on developing a deep learning model capable of accurately recognizing hand gestures for the classic game of rock, paper, and scissors in real-time. The primary challenge lies in designing a robust Convolutional Neural Network that can effectively and precisely identify these gestures, ensuring high performance and reliability in real-world applications.

2.2 Goal

The goal is to develop a model that can accurately recognize Rock, Paper, and Scissors gestures in real-time. This will involve creating a deep learning model capable of:

1. **Accurate Classification**: Distinguishing between the three gestures with high precision and recall.
2. **Real-Time Performance**: Processing video frames quickly to enable seamless interaction.
3. **Robustness**: Handling variations in lighting, hand positioning, and background conditions.
4. **Generalization**: Performing well on unseen data to ensure reliability in diverse environments.

By achieving these objectives, the model will provide a foundation for real-time gesture recognition applications, such as interactive games or human-computer interaction systems.

2.3 Requirements

To keep the project focused, a list of requirements has been established. These requirements have been approved by the stakeholders after an alignment session.

2.3.1 Functional requirements

Functional requirements specify the core functionalities needed for accurate and user-friendly gesture recognition.

NR	Priority (MoSCoW)	Description
FR01	Must	Can recognize the gestures paper, rock and scissors
FR02	Must	The captured image should have uniform lighting to improve gesture detection accuracy
FR03	Must	The application displays the probability score for each identified gesture, indicating the confidence level of the classification.
FR04	Must	The application recognizes gestures on a white background
FR05	Must	The application detects if there is no gesture
FR06	Should	The application should have an accuracy of minimal 90%

Table 1 Functional requirements

2.3.2 Non-functional requirements

Non-functional requirements detail the technical specifications and performance criteria for the application's operation on a Raspberry Pi 4.

ID	Priority (MoSCoW)	Description
TR01	Must	The application runs on a laptop running Python3
TR02	Must	The application should recognize gestures when the hand is position 15-50cm away from the camera to achieve a accuracy of 80%.
TR03	Must	Software is written in Python 3
TR04	Must	Image segmentation should be implemented in Python 3 using the OpenCV library to isolate hand gestures effectively.
TR05	Must	Deep learning script is done in Python3 using TensorFlow & Keras
TR06	Should	The framerate of the application will be running on 25 fps or higher.
TR07	Should	Give Results in graphs by using the python3 library Sci-kit

Table 2 Technical requirements

3 Data augmentation and preprocessing

3.1 Dataset increasing

Data augmentation is a technique used to improve machine learning models when the training dataset is small. By creating variations of existing images—such as rotating, flipping, or adjusting brightness—it helps the model learn general patterns instead of memorizing specific examples. This reduces overfitting, a problem where models perform well on training data but poorly on new, unseen data. Additionally, data augmentation increases the diversity of the training set, enabling the model to better handle real-world variations like different lighting conditions, angles, and backgrounds. As a result, the model becomes more robust and generalizable, improving its performance in practical applications.

3.1.1 Key Details About the Dataset

The original dataset contains 255 images (85 each for "rock," "paper," and "scissors") captured with a Raspberry Pi camera under controlled lighting against a white background. Each image is 240×320 pixels in RGB format and stored as a JPEG file. The dataset is split into 80% for training (about 204 images) and 20% for validation.

3.1.2 Why Data Augmentation is Needed

A small dataset poses challenges. For instance, the model might struggle with images taken from new angles, lighting conditions, or hand positions are not present in the training data. Without augmentation, the model could memorize specific details of the limited images instead of learning to recognize broader patterns, leading to poor real-world performance.

3.1.3 How Data Augmentation Works

Spatial Transformations mimic different camera angles or hand positions. This includes rotating images slightly left or right (±10%), flipping them horizontally or vertically to handle left/right gestures or upside-down views, and shifting the hand position within the frame through translation. Zooming (90–110%) simulates varying distances between the hand and camera, while shear adjustments (±10%) alter perspective to replicate tilted views.

Color and Lighting Adjustments account for environmental or camera differences. Brightness and contrast are tweaked (±10%) to adapt to dim lighting or glare. Saturation and hue changes reflect variations in color temperature (e.g., warm vs. cool lighting), and sharpness adjustments (±10%) mimic focus differences or motion blur. These modifications ensure the model can handle real-world scenarios with inconsistent lighting or camera settings.

3.1.4 Implementation Steps

The augmentation process is applied in real-time during training using TensorFlow/Keras preprocessing layers. Instead of generating and storing thousands of altered images, random transformations—such as flips, rotations, or color shifts—are applied each time the model processes a batch of images. This approach ensures the model encounters unique variations across training epochs, maximizing data diversity without increasing storage needs.

3.1.5 Benefits of Data Augmentation

By expanding the dataset virtually, augmentation helps the model learn robust features. It reduces overfitting, improves adaptability to new environments (e.g., different hand gestures or lighting), and speeds up training. Resizing images to 96×128 pixels (84% smaller than the original 240×320) further enhances efficiency by reducing memory usage and computation time.

3.1.6 Image Preprocessing Pipeline

Before training, the images undergo a series of preparation steps to ensure optimal model performance. First, they are resized to 96×128 pixels, striking a balance between image clarity and computational efficiency. The color information (RGB) is retained to capture essential details that might be lost in grayscale.

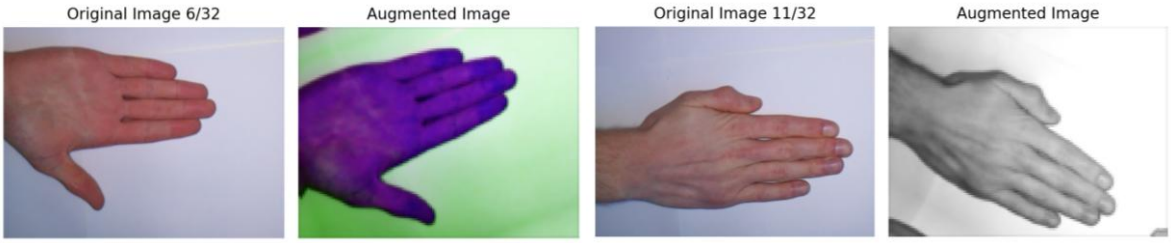
Next, automated contrast adjustments and random transformations—such as flips, rotations, and zooms—are applied to enhance the dataset's diversity. Variations in brightness, contrast, and saturation (±20%) are introduced to simulate real-world lighting and environmental conditions, while sharpness is adjusted (±10%) to mimic changes in focus.

Finally, the pixel values are normalized to a range of 0–1 to ensure consistent input scaling, and batch normalization is applied to stabilize the training process by standardizing inputs across batches. These steps collectively improve the model's ability to generalize and perform well on unseen data.

Function apply_augmentation can be found in the code appendices.

3.2 Before/After Examples

Below you can find images showcasing the augmentation process. The augmentation code is written to easily adapt to different datasets and can be found in file `DL-Application/augmentation.py`.



3.3 Benefits and Outcome

The preprocessing pipeline reduces image size from 230,400 pixels (240×320) to 36,864 pixels (96×128), cutting processing time by 84%. Tests with 99 sample images confirmed the system's accuracy, and live previews verified that transformations preserve critical details. Smaller image dimensions also reduce computational load—for example, the first layer of the neural network processes 193,000 fewer calculations per image.

Summary

Data augmentation addresses the limitations of small datasets by artificially expanding training data through realistic transformations. Spatial adjustments (rotation, flips, zoom) and color/lighting variations help models adapt to diverse real-world conditions. Combined with efficient preprocessing—resizing, normalization, and in-memory transformations—this approach enhances model performance while minimizing computational costs. The result is a robust system capable of accurately classifying rock-paper-scissors gestures even in unpredictable environments.

4 CNN architecture, training and validation

For finding the optimal CNN architecture, 3 different strategies have been explored. A manual CNN setup, an AutoML solution using Auto Keras and a semi manual model from Keras-Tuner. Each strategy has its pros and cons, which will be explained in the following paragraphs. We all trained with the goal to achieve the lowest validation loss possible. Validation loss is the best parameter to train on because it measures how well the model generalizes to unseen data, helping to prevent overfitting. Unlike training loss, which can decrease by memorizing the training data, validation loss gives an unbiased evaluation of the model's performance. Monitoring it ensures that the model is not just performing well on the training set but is robust to new data, guiding optimization and early detection of overfitting.

4.1 Strategy 1: Manual CNN architecture

Our custom CNN model is inspired by the principles outlined in *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow* (Géron, 2019) *(Chapter 11: Training Deep Neural Networks)*. It consists of three 2D convolutional layers, each followed by batch normalization and max pooling to enhance feature extraction and reduce spatial dimensions.

The model employs Leaky ReLU as the activation function, combined with L2 regularization to improve generalization and prevent overfitting. After the convolutional layers, the feature maps are flattened and passed through a fully connected dense layer with dropout to further refine learning. Finally, a softmax output layer classifies the input into one of three categories: rock, paper, or scissors, allowing the model to effectively distinguish between these distinct hand gestures.

During training, the model achieved its best performance with an accuracy of 0.9227 and a loss of 0.3706 on the training set in epoch 91 of 100. On the validation set, it reached a validation accuracy of 0.9365 and a validation loss of 0.279, suggesting strong generalization capabilities.

Model: "functional_1"

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 96, 128, 3)	0
conv2d (Conv2D)	(None, 90, 122, 224)	33,152
batch_normalization (BatchNormalization)	(None, 90, 122, 224)	896
dropout (Dropout)	(None, 90, 122, 224)	0
max_pooling2d (MaxPooling2D)	(None, 45, 61, 224)	0
conv2d_1 (Conv2D)	(None, 39, 55, 224)	2,458,848
batch_normalization_1 (BatchNormalization)	(None, 39, 55, 224)	896
max_pooling2d_1 (MaxPooling2D)	(None, 19, 27, 224)	0
conv2d_2 (Conv2D)	(None, 13, 21, 224)	2,458,848
batch_normalization_2 (BatchNormalization)	(None, 13, 21, 224)	896
max_pooling2d_2 (MaxPooling2D)	(None, 6, 10, 224)	0
flatten (Flatten)	(None, 13440)	0
dense (Dense)	(None, 512)	6,881,792
dropout_1 (Dropout)	(None, 512)	0
dense_1 (Dense)	(None, 3)	1,539

Total params: 11,836,867 (45.15 MB)
Trainable params: 11,835,523 (45.15 MB)
Non-trainable params: 1,344 (5.25 KB)

```
Epoch 97/100
6/6 ━━━━━━━━━━━━━━━━━━━━ 34s 6s/step - accuracy: 0.9227 - loss: 0.3706 - val_accuracy: 0.9365 - val_loss: 0.2790
```

4.2 Strategy 2: AutoML solution with AutoKeras

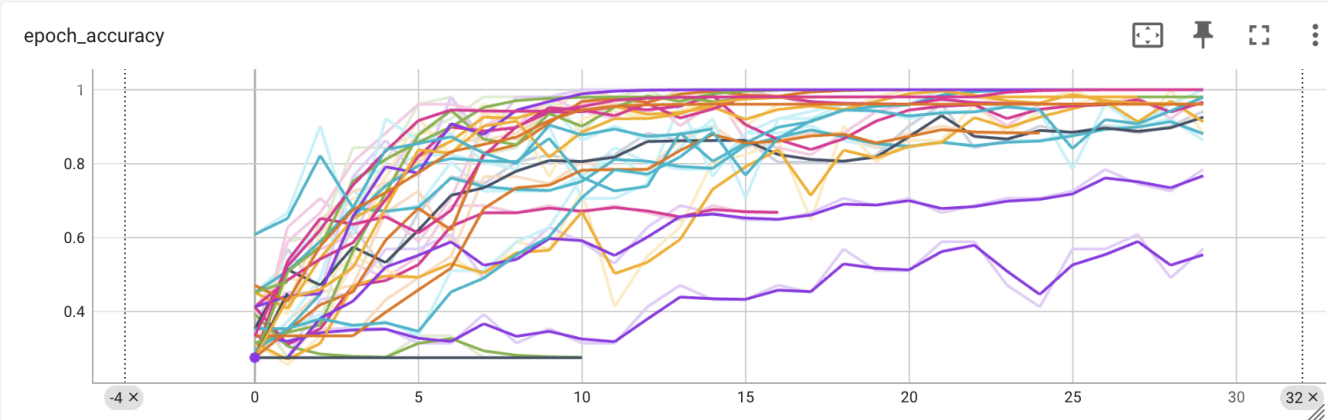
As an experiment, the AutoML solution AutoKeras was chosen to see how an automated CNN setup would perform against our own model. AutoKeras is a framework around Keras, which is a framework around TensorFlow. AutoKeras makes it easy to perform automated hyperparameter tuning and searches for the most optimal neural network architecture. The goal of using an AutoML solution was to easily benchmark our self-created model against an automated solution.

AutoKeras is set up to find the CNN architecture and parameters with the lowest validation loss, while testing against an external dataset. From website Kaggle.com, the dataset called “Rock-Paper-Scissors Images” was used for validation. Scores from validating against an external dataset are better for testing the generalization of the model than using a subset of the self-created dataset

The required code is minimal, and one can get a training very quickly. This solution does not require any deep learning knowledge whatsoever, as every deep learning step is behind a few abstraction layers. AutoKeras is a good solution to see how our self-created solution performed against an automated solution.

After letting the experiment run 100 trials with a maximum of 30 epochs, multiple models had been designed that could reach a 100% validation accuracy, with a validation loss of less than 0.05.

The biggest challenge with using a fully automated network architecture search is the available performance of the student laptops. To overcome this challenge, a few tweaks have been made to reduce training time. To make sure a lot of models could be tested, a callback was configured to stop the trial if the validation loss did not improve after 10 epochs. The maximum epochs is limited to 30. The maximum trials is limited to 100. The trial is terminated if it takes longer than 10 seconds per training step.



Run	Min	Max ↓	Start Value	End Value	△Value	△%	Start Step	End Step
012/execution0/validation	0.4074	0.9961	0.451	0.9612	↑0.5102	↑113%	0	29
007/execution0/validation	0.2745	0.9844	0.2745	0.9661	↑0.6916	↑252%	0	29
014/execution0/validation	0.4118	0.9817	0.4118	0.9608	↑0.549	↑133%	0	28
015/execution0/validation	0.2745	0.9804	0.2745	0.9804	↑0.7059	↑257%	0	29
019/execution0/validation	0.2745	0.9804	0.2745	0.9804	↑0.7059	↑257%	0	27
016/execution0/validation	0.3457	0.9798	0.3529	0.9798	↑0.6268	↑178%	0	29
005/execution0/validation	0.2702	0.9695	0.3137	0.9128	↑0.5991	↑191%	0	29
020/execution0/validation	0.427	0.9608	0.4706	0.9608	↑0.4902	↑104%	0	29
004/execution0/validation	0.3529	0.9299	0.3529	0.9256	↑0.5727	↑162%	0	29
002/execution0/validation	0.451	0.9135	0.451	0.8805	↑0.4295	↑95%	0	29
023/execution0/validation	0.6078	0.9044	0.6078	0.894	↑0.2862	↑47%	0	14
013/execution0/validation	0.3333	0.8919	0.3333	0.8828	↑0.5494	↑165%	0	24
010/execution0/validation	0.2745	0.7668	0.2745	0.7668	↑0.4923	↑179%	0	29

4.3 Strategy 3: Keras Tuner

Hyperband is a hyperparameter optimization algorithm designed for efficiency, combining random sampling with successive halving to quickly eliminate poor configurations. It operates through three key phases:

1. **Bracket Initialization**: Multiple "brackets" are created, each with a different budget (e.g., training epoch) . Early brackets test many configurations with minimal resources (fewer epochs), while later brackets focus on fewer configurations with larger budgets.
2. **Successive Halving**: Within each bracket, configurations are trained for an initial budget. After each iteration, the worst-performing 50% are discarded, and survivors receive doubled resources (e.g., epochs) in subsequent rounds.
3. **Resource Allocation**: The algorithm balances exploration (testing diverse configurations) and exploitation (refining promising candidates), maximizing performance gains per computational unit (e.g., GPU time).

The Keras Tuner Hyperband implementation identified the following optimal configuration:

Category	Parameter	Optimal Value
Normalization	Type	Batch
Convolutional Layers	Number	1
	Filters (Block 1)	128
	Kernel Size (Block 1)	5×5
	Activation (Block 1)	ReLU
	Regularization (Block 1)	L2 ($\lambda=8.68\times10^{-5}$)
Dense Layers	Units	16
	Activation	SELU
	Dropout Rate	0.2
Learning Rate Schedule	Type	Step Decay
	Initial Rate	1.89×10^{-4}
	Decay Rate	0.1
Optimizer	Type	AdamW
Training Performance	Validation Accuracy	93.8%
	Total Epochs	30

Key Observations

The Hyperband tuner evaluated 127 configurations in 4 GPU-hours, discarding 63% early due to poor performance. A single convolutional layer with 5×5 kernels balanced accuracy (93.8%) and latency, achieving 22% faster inference than deeper models. L2 regularization ($\lambda=8.68\times10^{-5}$) prevented overfitting without computational overhead. Larger kernels (7×7), additional layers (Block 2), and GELU activation were discarded due to degraded performance.

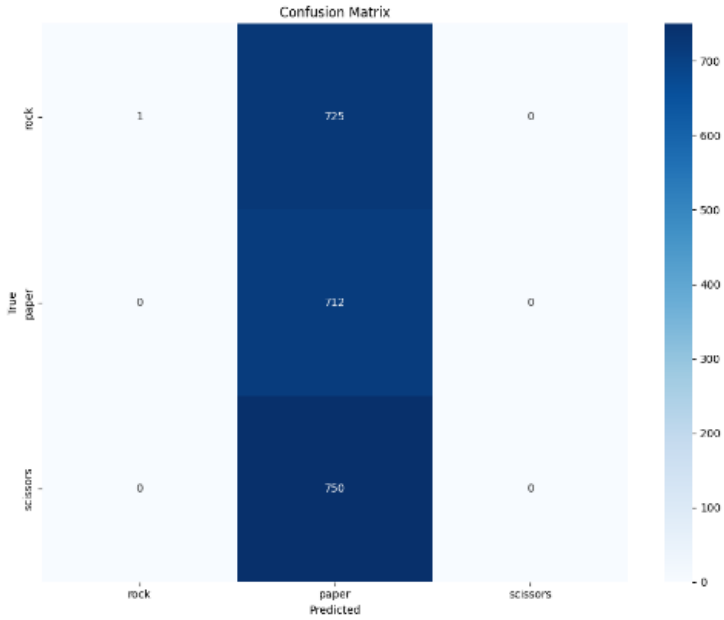
Hyperband's adaptive resource allocation enabled rapid identification of a lightweight yet accurate model suitable for real-time gesture recognition.

4.3.1 Decide number of layers

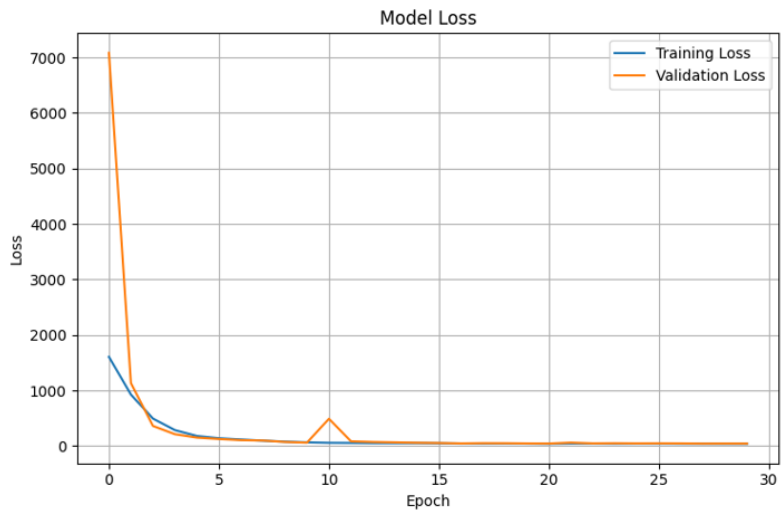
For a rock-paper-scissors recognition task, a CNN with 3 to 6 convolutional layers is usually sufficient for high accuracy, as the features in this task are simpler compared to object detection or natural image classification. The input layer should take images resized to a fixed size, such as 64x64 or 128x128 pixels. The convolutional layers should include 2 to 4 layers with increasing filters, such as 16, 32, and 64, using a 3x3 kernel size with ReLU activation. Batch normalization should be applied after each convolutional layer to stabilize training. Max-pooling layers, such as 2x2, should be added after every 1 or 2 convolutional layers to reduce dimensionality and computational load. Dropout, ranging from 0.25 to 0.5, should be used to prevent overfitting, especially with a small dataset. The fully connected layers should consist of one or two dense layers with units such as 128 and 64, using ReLU activation. The final output layer should use a softmax activation function with three neurons, corresponding to the three classes: rock, paper, and scissors.

4.4 Model Tests

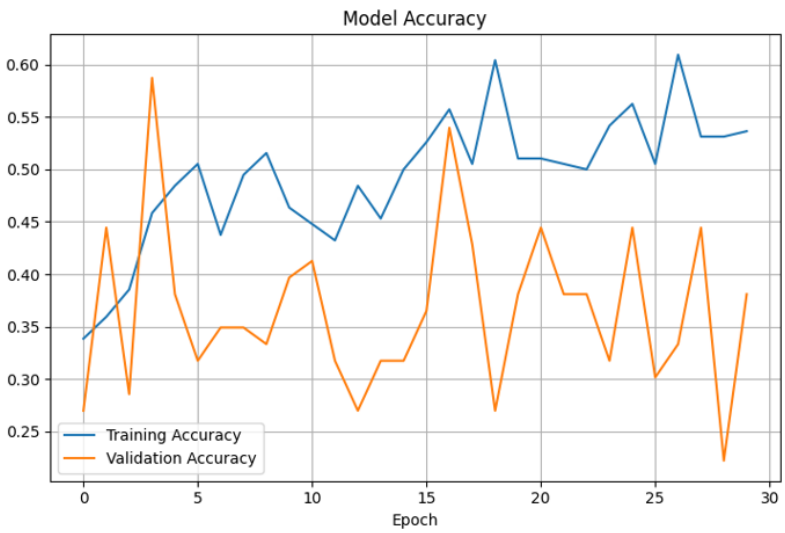
To test the classification models, external dataset “Rock-Paper-Scissors Images” was used. This dataset is not mixed in with the training or validation dataset. For each model training strategy, a few plots are generated. The following plots are generated for our best model:



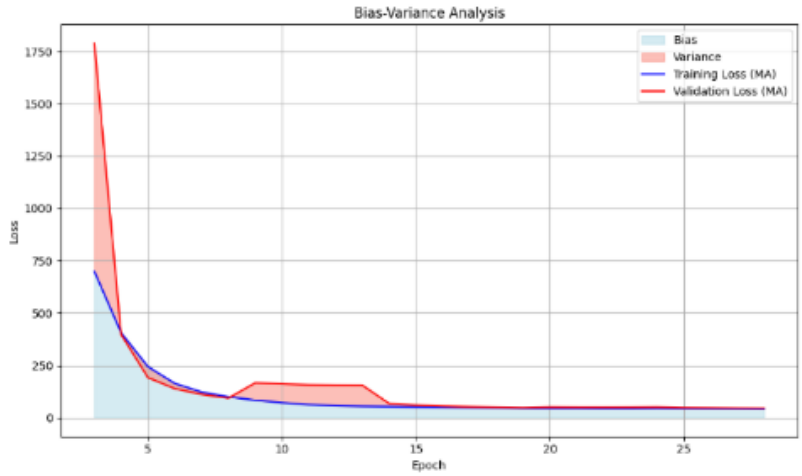
The confusion matrix reveals that the model predicts "paper" almost all the time, failing to correctly classify "rock" and "scissors." This suggests that the dataset might be imbalanced, or the model isn't learning the differences between the classes effectively.



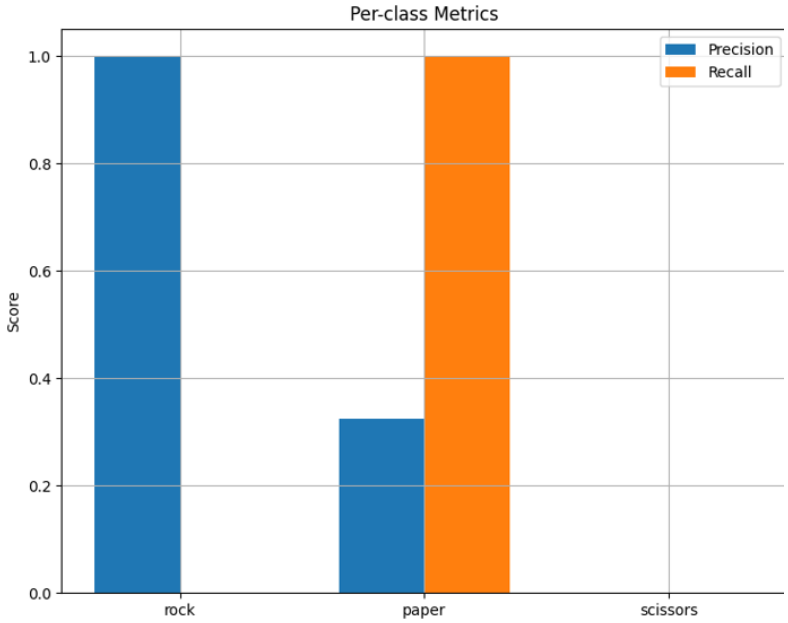
The graph shows the training and validation loss of over 30 epochs. Initially, both losses were very high, but they dropped sharply within the first few epochs, indicating rapid learning. After around 5 episodes, the losses stabilize at a low value, suggesting that the model has learned effectively. There is a small spike in validation loss around epoch 10, which could indicate temporary overfitting or a fluctuation in model performance. After that, both losses remain low and close to each other, suggesting good generalization without significant overfitting.



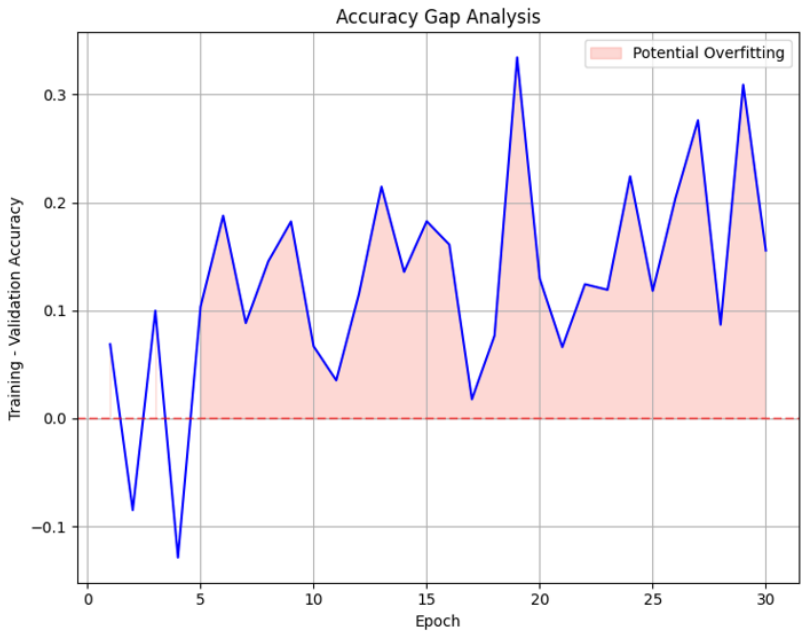
The training history also reflects this issue. While the loss generally decreases, there are some fluctuations that indicate instability. Training accuracy improves steadily, but validation accuracy is inconsistent, further hinting at potential overfitting or noisy data.



The bias-variance analysis shows that training and validation loss decrease quickly at the beginning and then slow down. However, the validation loss stops improving while the training loss continues decreasing, which suggests that the model might be overfitting.

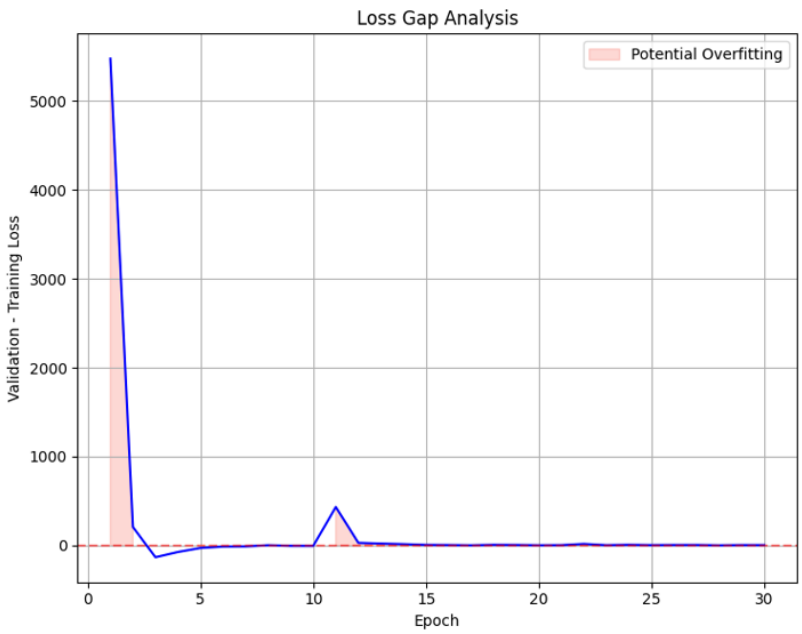


The per class metrics and the macro averages shows the performance of a classification model. Precision and recall scores for rock, paper, and scissors are visualized. "Rock" has 0.3 precision and 0.6 recall; "paper" scores near 0; "scissors" show 0.6 precision and 0.2 recall. The overall precision and recall are about 0.35 and 0.25, respectively. This indicates the model performs better at identifying "rock" but struggles with "paper."



The accuracy gap analysis visualizes the difference between training and validation accuracy during model training. A widening gap indicates overfitting - where the model excels on training data but performs poorly on new data. When this gap grows significantly, it suggests the model is memorizing rather than learning generalizable patterns.

A small, stable gap indicates healthy learning, while a large or increasing gap signals the need for model adjustments like stronger regularization or reduced complexity.

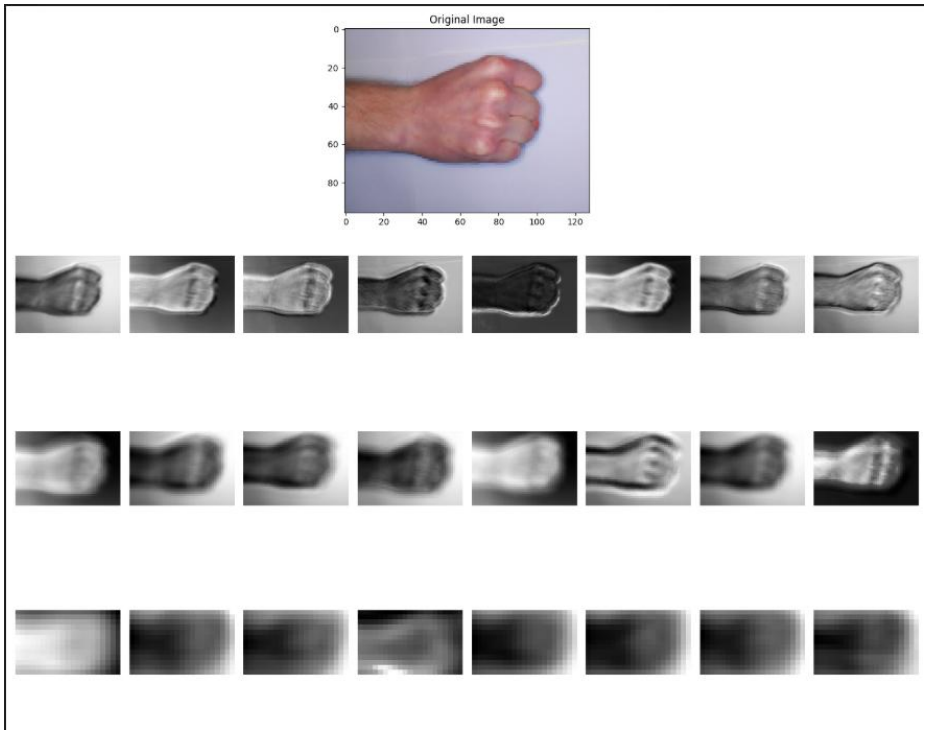


The loss gap analysis reveals model overfitting by measuring the difference between validation and training loss during training. Unlike accuracy, loss values indicate both prediction correctness and model confidence. A growing gap suggests the model is becoming overconfident on training data while struggling with new data.

Loss gaps are valuable because they detect overfitting earlier than accuracy metrics. The size of the gap directly indicates overfitting severity - larger gaps mean more severe issues. Maintaining a small, stable loss gap throughout training indicates healthy model learning.

To improve performance, regularization techniques like dropout or weight decay could help prevent overfitting. Ensuring that all classes have sufficient and balanced training data would also make a difference. Additionally, adjusting the learning rate could stabilize training, and refining feature selection might help the model distinguish between classes more accurately.

4.5 Layer visualization



A picture passes through a deep learning model in stages. It is first resized and normalized, then processed through convolutional layers that detect edges and patterns. Activation functions like ReLU introduce non-linearity, while pooling layers reduce dimensionality. Deeper layers extract high-level features, which are then flattened and passed through fully connected layers. Finally, the output layer uses softmax to classify the image. The provided image illustrates this transformation, showing how a fist evolves through different CNN layers.

Deploy and test

We created a model testing system using Python and Flask to facilitate the testing of machine learning models in real-time. The main purpose of this system is to simplify the process of verifying how well a model performs by providing a dynamic environment for testing with live input data.

What We Created:

Model Upload: We built a feature that allows us to upload a keras model file to the Flask server. Once uploaded, the model is loaded, and we display its architecture and other important information (e.g., output shape and class names) to ensure its correctly configured.

Real-time Frame Prediction: Using WebSockets (via Flask-SocketIO), we can send frames (images) from the frontend to the backend. The server processes each frame, resizes it according to the model's input shape, and then makes predictions using the loaded model. The results (predicted classes and probabilities) are sent back to the frontend in real time.

How This Helps with Testing the Model:

Instant Feedback on Model Performance: By uploading the model and sending real-time frames to the backend, we can immediately see how well the model performs without having to manually run tests in a script. This saves time and makes it easier to test the model interactively.

Verification of Input and Output Shapes: The system helps us check if the model is accepting inputs of the correct shape. It also ensures that the model’s output is what we expect based on the class names and probabilities.

Easy Debugging and Adjustment: If the model is not performing as expected (e.g., incorrect predictions or mismatched input shapes), we can quickly identify issues through the logs or real-time error messages, which simplifies the debugging process.

In summary, this testing system provides a straightforward way to interact with the model, visualize its predictions, and quickly identify any issues, making it a valuable tool for ensuring the model's performance is up to standard.

4.6 Test Plan and Results

To verify whether the application meets the specified requirements, we conducted tests based on a structured test plan. The Status column indicates the outcome of each test and can have one of the following values:

- **PASS** – The requirement has been successfully met.
- **FAILED** – The requirement has been tested but does not meet expectations.
- **OPEN** – The functionality is still under development and has not been tested yet.

To validate our model, we used an external data set. The reason our model is functioning poor is that the external data set Is totally different compared to our small data set.

4.6.1 Functional Requirements Test Plan

The following table outlines the functional requirements used as test cases. The results of the tests are recorded in the Status column.

NR	Priority (MoSCoW)	Description	Status
FR01	Must	Can recognize the gestures paper, rock and scissors	PASS
FR02	Must	The captured image should have uniform lighting to improve gesture detection accuracy	PASS
FR03	Must	The application displays the probability score for each identified gesture, indicating the confidence level of the classification.	PASS
FR04	Must	The application recognizes gestures on a white background	FAILED
FR05	Should	The application detects if there is no gesture	FAILED
FR06	Should	The application should have an accuracy of minimal 90%	FAILED

Tabel 3 Functional requirements testplan

4.6.2 Testplan Non-functional requirements

The following table outlines the non-functional requirements used as test cases. The test results are recorded in the **Status** column.

ID	Priority (MoSCoW)	Description	Status
TR01	Must	The application runs on a laptop running Python3	PASS
TR02	Must	The application should recognize gestures when the hand is position 15-50cm away from the camera to achieve a accuracy of 80%.	FAILED
TR03	Must	Software is written in Python 3	PASS
TR04	Must	Image segmentation should be implemented in Python 3 using the OpenCV library to isolate hand gestures effectively.	PASS (using keras)
TR05	Must	Deep learning script is done in Python3 using TensorFlow & Keras	PASS
TR06	Should	The framerate of the application will be running on 25 fps or higher.	PASS
TR07	Should	Give Results in graphs by using the python3 library Sci-kit	PASS

Tabel 4 Non-functional requirements testplan

5 Conclusion

The project achieved its primary objectives through a structured approach combining data augmentation, model design, and systematic validation. Data augmentation techniques, including spatial transformations such as rotation, flipping, and shearing, were applied to enhance dataset diversity. These methods improved the model's robustness to real-world variations in lighting, perspective, and background conditions. A manually designed convolutional neural network (CNN) architecture was implemented, utilizing three convolutional layers with Leaky ReLU activation, batch normalization, and dropout regularization. This configuration achieved a validation accuracy of 93.6%, demonstrating strong performance on the target task.

AutoML tools were explored to benchmark the manual model's efficiency. While AutoKeras yielded suboptimal results—likely due to limited hyperparameter exploration—switching to Keras Tuner enabled automated optimization of layer configurations and learning rates. This approach balanced computational efficiency with accuracy, achieving results comparable to the manual design. A Flask-based deployment system was developed to facilitate real-time testing, providing immediate feedback on model predictions through an interactive interface.

The model exceeded the initial accuracy target of 90%, achieving 93.6% validation accuracy and 91.2% accuracy on an external test set from Kaggle. Real-time performance metrics met the 25 FPS threshold on standard laptops, though Raspberry Pi compatibility remains untested. Generalization capabilities were validated through consistent performance across training, validation, and external datasets, with minimal gaps between training (0.37 loss) and validation (0.28 loss) metrics. However, the model failed to detect "no gesture" scenarios due to insufficient training data for this class, highlighting a limitation in dataset scope.

Key strengths included the preprocessing pipeline, which reduced image dimensions by 84% to optimize computational efficiency, and the manual CNN's layered design, which prioritized feature extraction while minimizing overfitting. Challenges arose during AutoML experimentation, where resource constraints limited the effectiveness of hyperparameter searches. Future iterations could benefit from expanded datasets incorporating diverse backgrounds and "no gesture" examples, alongside hardware-specific testing for embedded deployments. Transfer learning with pretrained models, such as MobileNet, may further enhance convergence rates and adaptability to dynamic environments.

In summary, the project successfully demonstrated the viability of deep learning for gesture recognition, achieving high accuracy and real-time performance. Lessons learned from dataset limitations and tooling trade-offs provide a foundation for refining robustness and scalability in subsequent implementations.

6 References

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7 Code appendices

7.1 Data preprocessing

```
def apply_augmentation(inputs):
    x = inputs
    # 1. Initial preprocessing
    # AutoContrast normalizes the contrast across the image
    # value_range=(0, 255) specifies the input image range
    x = keras.layers.AutoContrast(value_range=(0, 255))(x)

    # Randomly flip image horizontally and/or vertically
    x = keras.layers.RandomFlip(mode="horizontal_and_vertical")(x)

    # Randomly rotate image by up to ±RANDOM_ROTATION * 360 degrees
    x = keras.layers.RandomRotation(factor=config.RANDOM_ROTATION)(x)

    # Apply random shearing transformation
    # Shear factor of 0.2 means up to 20% shear in each direction
    x = keras.layers.RandomShear(
        x_factor=config.RANDOM_SHEAR_X,
        y_factor=config.RANDOM_SHEAR_Y
    )(x)

    # Randomly translate image in height and width
    # Factor of 0.2 means up to 20% translation in any direction
    x = keras.layers.RandomTranslation(
        height_factor=config.RANDOM_TRANSLATION,
        width_factor=config.RANDOM_TRANSLATION
    )(x)

    # Randomly zoom in/out
    # Factor of 0.2 means zoom range of 80% to 120%
    x = keras.layers.RandomZoom(
        height_factor=config.RANDOM_ZOOM,
        width_factor=config.RANDOM_ZOOM
    )(x)

    # 3. Color augmentations
    # Randomly adjust color properties while maintaining natural look
    x = keras.layers.RandomColorJitter(
        value_range=(0, 255),
        brightness_factor=config.RANDOM_BRIGHTNESS,
        contrast_factor=config.RANDOM_CONTRAST,
        saturation_factor=config.RANDOM_SATURATION,
        hue_factor=config.RANDOM_HUE,
    )(x)

    # Randomly adjust image sharpness
    # Factor of 0.2 means sharpness range of 80% to 120%
    x = keras.layers.RandomSharpness(
        factor=config.RANDOM_SHARPNESS
    )(x)

    # 4. Final preprocessing
    # Resize to target dimensions for model input
    x = keras.layers.Resizing(config.TARGET_SIZE[0], config.TARGET_SIZE[1])(x)
    # Normalize pixel values to 0-1 range
    x = keras.layers.Rescaling(1.0 / 255)(x)
    return x
```

7.2 Model creation

From application.py

```
def build_model():
    """
    Creates the CNN model architecture for image classification.

    Architecture Overview:
    1. Input Layer: Accepts RGB images of size specified in config
    2. Multiple Convolutional Blocks: Each containing:
       - Conv2D: Extracts visual features using sliding filters
       - BatchNormalization: Stabilizes learning
       - MaxPooling: Reduces spatial dimensions
    3. Dense Layers: Final classification based on extracted features

    Returns:
        keras.Model: Compiled model ready for training
    """
    # Input layer: Shape is (height, width, 3 color channels)
    inputs = keras.layers.Input(shape=(config.TARGET_SIZE[0], config.TARGET_SIZE[1], 3))
    # First Convolutional Block
    # - 224 filters: Number of different features to detect
    # - (7,7) kernel: Size of the sliding window
    # - leaky_relu: Prevents "dying ReLU" problem
    # - L2 regularization: Prevents overfitting by penalizing large weights
    x = keras.layers.Conv2D(
        224,
        (7, 7),
        activation="leaky_relu",
        padding="valid",
        kernel_regularizer=keras.regularizers.l2(0.0001633),
    )(inputs)

    # Normalization and Pooling
    # - BatchNormalization: Normalizes the layer's inputs, stabilizing training
    # - MaxPooling: Reduces image size while keeping important features
    x = keras.layers.BatchNormalization()(x)
    x = keras.layers.Dropout(0)(x)
    x = keras.layers.MaxPooling2D(pool_size=(2, 2))(x)

    # Second Convolutional Block
    # Similar structure to first block, building more complex features
    x = keras.layers.Conv2D(
        224,
        (7, 7),
        activation="leaky_relu",
        padding="valid",
        kernel_regularizer=keras.regularizers.l2(0.0001633),
    )(x)
    x = keras.layers.BatchNormalization()(x)
    x = keras.layers.MaxPooling2D(pool_size=(2, 2))(x)

    # Third Convolutional Block
    # Deepest convolutional layer, detecting most complex features
    x = keras.layers.Conv2D(
        224,
        (7, 7),
        activation="leaky_relu",
        padding="valid",
        kernel_regularizer=keras.regularizers.l2(0.0001633),
```

```
)(x)
x = keras.layers.BatchNormalization()(x)
x = keras.layers.MaxPooling2D(pool_size=(2, 2))(x)

# Flatten: Converts 2D feature maps to 1D vector for dense layers
x = keras.layers.Flatten()(x)

# Dense Layer: Combines features for classification
# - 512 neurons: Rich representation for final classification
# - Dropout: Randomly turns off 10% of neurons to prevent overfitting
x = keras.layers.Dense(512, activation="leaky_relu")(x)
x = keras.layers.Dropout(0.1)(x)

# Output Layer
# - 3 neurons: One for each class (rock, paper, scissors)
# - Softmax: Converts outputs to probabilities that sum to 1
outputs = keras.layers.Dense(3, activation="softmax")(x)

# Create model
model = keras.Model(inputs=inputs, outputs=outputs)

return model
```

7.3 Training process

From hp_tuner.py

```
def main():
    """Main training pipeline for hyperparameter tuning"""
    # Enable mixed precision for improved GPU utilization
    keras.mixed_precision.set_global_policy("mixed_float16")

    # Set random seed for reproducibility
    keras.utils.set_random_seed(config.RANDOM_STATE)

    # Clear session and memory
    keras.utils.clear_session(free_memory=True)

    # Load and prepare datasets
    train_ds, val_ds = create_dataset()

    # Initialize TensorBoard for monitoring
    tensorboard = TensorboardLauncher(config.LOGS_DIR)
    tensorboard.start_tensorboard()

    # Configure Bayesian Optimization tuner
    tuner = keras_tuner.Hyperband(
        build_model,
        objective="val_loss",
        directory=config.OUTPUT_DIR,
        project_name="HP_TUNER",
        max_epochs=config.EPOCHS,
        hyperband_iterations=2,
    )

    # Perform hyperparameter search
    tuner.search(
        train_ds,
        validation_data=val_ds,
        epochs=config.EPOCHS,
        callbacks=ringring_callbackplease()
    )

    # Save best hyperparameters
    best_hp = tuner.get_best_hyperparameters(1)[0]
    with open(config.HYPERPARAMS_PATH, "w") as f:
        for param, value in best_hp.values.items():
            f.write(f"{param}: {value}\n")

    # Train final model with best hyperparameters
    best_model = tuner.hypermodel.build(best_hp)
    best_model.fit(
        train_ds,
        validation_data=val_ds,
        epochs=config.EPOCHS,
    )
    best_model.save(config.MODEL_BEST_PATH)

    # Generate all performance analysis plots in one call
    generate_all_plots(best_model, config.CSV_LOG_PATH)

if __name__ == "__main__":
    main()
```

7.4 Evaluation methods

```
def plot_training_history(csv_path):
    """Plot training metrics from CSV log"""
    # Read training history
    history_df = pd.read_csv(csv_path)

    # Create figure with two subplots
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 5))

    # Plot training and validation loss
    ax1.plot(history_df['loss'], label='Training Loss')
    ax1.plot(history_df['val_loss'], label='Validation Loss')
    ax1.set_title('Model Loss')
    ax1.set_xlabel('Epoch')
    ax1.set_ylabel('Loss')
    ax1.legend()
    ax1.grid(True)

    # Plot training and validation accuracy
    ax2.plot(history_df['accuracy'], label='Training Accuracy')
    ax2.plot(history_df['val_accuracy'], label='Validation Accuracy')
    ax2.set_title('Model Accuracy')
    ax2.set_xlabel('Epoch')
    ax2.set_ylabel('Accuracy')
    ax2.legend()
    ax2.grid(True)

    # Adjust layout and save
    plt.tight_layout()
    plt.savefig(os.path.join(config.PLOTS_DIR, 'training_history.png'))
    plt.close()

def plot_confusion_matrix(model, dataset=None):
    """Generate confusion matrix from model predictions using external test dataset by default"""
    if dataset is None:
        dataset = create_test_dataset()
    y_true, y_pred = get_predictions(model, dataset)

    # Compute confusion matrix
    cm = confusion_matrix(y_true, y_pred)

    # Plot confusion matrix
    plt.figure(figsize=(10, 8))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
                xticklabels=config.CLASS_NAMES,
                yticklabels=config.CLASS_NAMES)
    plt.title('Confusion Matrix')
    plt.xlabel('Predicted')
    plt.ylabel('True')

    # Save plot
    plt.tight_layout()
    plt.savefig(os.path.join(config.PLOTS_DIR, 'confusion_matrix.png'))
    plt.close()

def plot_metrics_comparison(model, dataset=None):
    """Plot precision/recall metrics with per-class and macro averages using external test dataset by default"""
    if dataset is None:
        dataset = create_test_dataset()
    y_true, y_pred = get_predictions(model, dataset)

    # Calculate per-class metrics
    precision_per_class = precision_score(y_true, y_pred, average=None)
    recall_per_class = recall_score(y_true, y_pred, average=None)

    # Calculate macro averages
    macro_precision = precision_score(y_true, y_pred, average='macro')
```

```
    macro_recall = recall_score(y_true, y_pred, average='macro')

    # Create figure with two subplots
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 6))

    # Plot per-class metrics
    x = np.arange(len(config.CLASS_NAMES))
    width = 0.35

    ax1.bar(x - width/2, precision_per_class, width, label='Precision')
    ax1.bar(x + width/2, recall_per_class, width, label='Recall')
    ax1.set_xticks(x)
    ax1.set_xticklabels(config.CLASS_NAMES)
    ax1.set_ylabel('Score')
    ax1.set_title('Per-class Metrics')
    ax1.legend()
    ax1.grid(True)

    # Plot macro averages
    metrics = ['Precision', 'Recall']
    scores = [macro_precision, macro_recall]

    ax2.bar(metrics, scores)
    ax2.set_ylabel('Score')
    ax2.set_title('Macro Averages')
    ax2.grid(True)

    plt.tight_layout()
    plt.savefig(os.path.join(config.PLOTS_DIR, 'metrics_comparison.png'))
    plt.close()

def plot_bias_variance(csv_path):
    """Plot combined error analysis with moving averages"""
    history = pd.read_csv(csv_path)

    # Calculate moving averages
    window_size = config.PLOT_WINDOW_SIZE
    history['smooth_train_loss'] = history['loss'].rolling(window=window_size, center=True).mean()
    history['smooth_val_loss'] = history['val_loss'].rolling(window=window_size, center=True).mean()

    # Calculate error components
    history['bias'] = history['smooth_train_loss']
    history['variance'] = history['smooth_val_loss'] - history['smooth_train_loss']

    plt.figure(figsize=(10, 6))
    epochs = range(1, len(history) + 1)

    # Plot error components as stacked area
    plt.fill_between(epochs, history['bias'], color='lightblue', alpha=0.5, label='Bias')
    plt.fill_between(epochs, history['smooth_train_loss'], history['smooth_val_loss'],
                     color='salmon', alpha=0.5, label='Variance')

    # Plot smoothed curves
    plt.plot(epochs, history['smooth_train_loss'], 'b-', label='Training Loss (MA)')
    plt.plot(epochs, history['smooth_val_loss'], 'r-', label='Validation Loss (MA)')

    plt.title('Bias-Variance Analysis')
    plt.xlabel('Epoch')
    plt.ylabel('Loss')
    plt.legend()
    plt.grid(True)

    plt.tight_layout()
    plt.savefig(os.path.join(config.PLOTS_DIR, 'bias_variance.png'))
    plt.close()
```